

A330 Limitations, know-cold set, FCOM R17 LIM

254 limitations. Number, units, condition. ha330pilot.app · FCOM R17 (15 MAY 26) · study aid only, the current manual governs

Limitation	Value	Ref
APU		
APU start attempts	Wait 60 min after three consecutive start attempts	LIM-APU-10
APU maximum rotor speed (N), APU running	107 %	LIM-APU-10
Maximum EGT for APU start	1 250 °C	LIM-APU-10
Maximum EGT for APU running	650 °C	LIM-APU-10
APU maximum operating altitude with JET B or JP4, When JET B or JP4 is used	25 000 ft	LIM-APU-20
APU bleed maximum altitude to assist engine start, APU bleed for engine start	20 000 ft	LIM-APU-20
APU bleed maximum altitude for air conditioning and pressurization, single pack, Single pack operation	22 500 ft	LIM-APU-20
APU bleed maximum altitude for air conditioning and pressurization, dual pack, Dual pack operation	17 500 ft	LIM-APU-20
APU start attempts	Wait 60 min after three consecutive start attempts	A330F LIM-APU-10
APU maximum rotor speed (N), APU running	107 %	A330F LIM-APU-10
Maximum EGT for APU start	1 250 °C	A330F LIM-APU-10
Maximum EGT for APU running	650 °C	A330F LIM-APU-10
APU maximum operating altitude with JET B or JP4, When JET B or JP4 is used	25 000 ft	A330F LIM-APU-20
APU bleed maximum altitude to assist engine start, APU bleed for engine start	20 000 ft	A330F LIM-APU-20
APU bleed maximum altitude for air conditioning and pressurization, single pack, Single pack operation	22 500 ft	A330F LIM-APU-20
APU bleed maximum altitude for air conditioning and pressurization, dual pack, Dual pack operation	17 500 ft	A330F LIM-APU-20
Air Bleed/Cond/Press/Vent		
Time without air conditioning supply, With passengers on board	20 min (not recommended to exceed)	LIM-AIR
Maximum positive differential pressure, Cabin pressure	9.25 PSI	LIM-AIR
Maximum negative differential pressure, Cabin pressure	-0.73 PSI	LIM-AIR
Safety relief valve setting, Cabin pressure	8.85 PSI	LIM-AIR
Differential pressure and safety valve setting tolerance, Applies to max delta-p and safety valve setting	± 7 hPa (0.1 PSI)	LIM-AIR
Maximum normal cabin altitude, Depending on CPC standard, for flights longer than 2.5 h	7 460 ft	LIM-AIR
Time without air conditioning supply, With passengers on board	20 min (not recommended to exceed)	A330F LIM-AIR
Maximum positive differential pressure, Cabin pressure	9.25 PSI	A330F LIM-AIR
Maximum negative differential pressure, Cabin pressure	-0.73 PSI	A330F LIM-AIR
Safety relief valve setting, Cabin pressure	8.85 PSI	A330F LIM-AIR
Differential pressure and safety valve setting tolerance, Applies to max delta-p and safety valve setting	± 7 hPa (0.1 PSI)	A330F LIM-AIR
Maximum normal cabin altitude, Depending on CPC standard, for flights longer than 2.5 h	7 460 ft	A330F LIM-AIR
Aircraft General		
Minimum TAT, Environmental envelope	-53 °C	LIM-AG-OPS-ENV
Runway slope (mean), Airport operations	±2 %	LIM-AG-OPS-ARPT_WIND
Nominal runway width, Airport operations	148 ft/45 m	LIM-AG-OPS-ARPT_WIND
Maximum certified crosswind for takeoff, Takeoff, A330 PAX fleet; AFM limitation, engine limitation	32 kt (gust included)	LIM-AG-OPS-ARPT_WIND
Maximum demonstrated crosswind for landing, Landing, A330 PAX fleet; demonstrated value, not an AFM limitation	45 kt (gust included)	LIM-AG-OPS-ARPT_WIND
Maximum tailwind for takeoff	15 kt	LIM-AG-OPS-ARPT_WIND
Maximum tailwind for landing	10 kt	LIM-AG-OPS-ARPT_WIND

Limitation	Value	Ref
Maximum wind for passenger door operation, Passenger door operation on ground	40 kt (or 50 kt if the aircraft nose is into the wind)	LIM-AG-OPS-ARPT_WIND
Maximum wind for FWD and AFT cargo door operation, FWD and AFT cargo door operation on ground	40 kt (or 50 kt if the aircraft nose is into the wind, or if the doors are on the downwind side)	LIM-AG-OPS-ARPT_WIND
Wind speed by which passenger, FWD and AFT cargo doors must be closed, Doors must be closed before the wind speed exceeds this value	60 kt	LIM-AG-OPS-ARPT_WIND
Maximum operating altitude, slats and flaps retracted	41 450 ft	AFM LIM-GEN
Minimum TAT, Environmental envelope	-53 °C	A330F LIM-AG-OPS-ENV
Runway slope (mean), Airport operations	±2 %	A330F LIM-AG-OPS-ARPT_WIND
Nominal runway width, Airport operations	148 ft/45 m	A330F LIM-AG-OPS-ARPT_WIND
Maximum certified crosswind for takeoff, Takeoff, A330 freighter fleet; AFM limitation, engine limitation	32 kt (gust included)	A330F LIM-AG-OPS-ARPT_WIND
Maximum demonstrated crosswind for landing, Landing, A330 freighter fleet; demonstrated value, not an AFM limitation	40 kt (gust included)	A330F LIM-AG-OPS-ARPT_WIND
Maximum tailwind for takeoff	15 kt	A330F LIM-AG-OPS-ARPT_WIND
Maximum tailwind for landing	10 kt	A330F LIM-AG-OPS-ARPT_WIND
Maximum wind for passenger door operation, Passenger door operation on ground	40 kt (or 50 kt if the aircraft nose is into the wind)	A330F LIM-AG-OPS-ARPT_WIND
Maximum wind for FWD and AFT cargo door operation, FWD and AFT cargo door operation on ground	40 kt (or 50 kt if the aircraft nose is into the wind, or if the doors are on the downwind side)	A330F LIM-AG-OPS-ARPT_WIND
Wind speed by which passenger, FWD and AFT cargo doors must be closed, Doors must be closed before the wind speed exceeds this value	60 kt	A330F LIM-AG-OPS-ARPT_WIND
Maximum wind for main deck cargo door operation, Main deck cargo door operation on ground	40 kt (no into-wind increase)	A330F LIM-AG-OPS-ARPT_WIND
Wind speed by which the main deck cargo door must be closed	60 kt	A330F LIM-AG-OPS-ARPT_WIND
Automatic Flight		
Minimum AP engagement height at takeoff	100 ft AGL and at least 5 s after liftoff	LIM-AFS-10-10
Minimum AP use height in approach with F-G/S mode	200 ft AGL	LIM-AFS-10-10
Minimum AP use height in approach with FINAL APP, V/S or FPA mode	250 ft AGL	LIM-AFS-10-10
Minimum AP use height in circling approach	500 ft AGL	LIM-AFS-10-10
Minimum AP use height, ILS approach with CAT1 on FMA, ILS approach when CAT1 is displayed on the FMA	160 ft AGL	LIM-AFS-10-10
Minimum AP use height, GLS approach without AUTOLAND on FMA, GLS approach when AUTOLAND is not displayed on the FMA	160 ft AGL	LIM-AFS-10-10
Minimum AP use height, LPV approach with SLS when CAT1 on FMA, LPV approach (with SLS) when CAT1 is displayed on the FMA	160 ft AGL	LIM-AFS-10-10
Minimum AP use height, ILS approach with CAT2 or CAT3 on FMA	0 ft AGL if autoland	LIM-AFS-10-10
Minimum AP engagement height after a manual go-around	100 ft AGL	LIM-AFS-10-10
Minimum AP use height in all other phases	500 ft AGL	LIM-AFS-10-10
AP/FD in OP DES or DES mode in approach, Approach using OP DES or DES mode	FCU selected altitude set to or above the higher of MDA/MDH or 500 ft AGL	LIM-AFS-10-10
RNP AR achievable level	Could be below 0.3 NM, depending on the RNP AR application (refer to ACD)	LIM-AFS-10-20
ILS CAT II minimum decision height	100 ft	LIM-AFS-20
AP disconnect height for manual landing, ILS CAT II, ILS CAT II manual landing	No later than 80 ft AGL	LIM-AFS-20
SA CAT II minimum decision height	100 ft	LIM-AFS-20
AP disconnect height, SA CAT II automatic approach without autoland	No later than 80 ft AGL	LIM-AFS-20
OTS CAT II minimum decision height	100 ft	LIM-AFS-20

Limitation	Value	Ref
AP disconnect height, OTS CAT II automatic approach without autoland	No later than 80 ft AGL	LIM-AFS-20
ILS CAT III fail passive (single) minimum decision height	50 ft	LIM-AFS-20
ILS CAT III fail operational (dual) alert height	200 ft	LIM-AFS-20
Minimum runway visual range, CAT III without DH	75 m	LIM-AFS-20
CAT II / CAT III autoland with engine out, Engine-out CAT II/CAT III autoland	Approved only in Configuration 3, engine-out procedures completed before 1 000 ft in approach	LIM-AFS-20
Maximum headwind for ILS CAT II/CAT III and GLS CAT I	35 kt	LIM-AFS-20
Maximum tailwind for ILS CAT II/CAT III and GLS CAT I	10 kt	LIM-AFS-20
Maximum crosswind for ILS CAT II/CAT III and GLS CAT I	15 kt	LIM-AFS-20
Automatic landing approved configurations, ILS CAT II/CAT III autoland and GLS CAT I autoland	CONF 3 and CONF FULL	LIM-AFS-20-10
Automatic landing demonstrated glide slope angle, Automatic landing demonstration envelope	Between -2.5 ° and -3.25 °	LIM-AFS-20-10
Automatic landing maximum airfield elevation, Automatic landing demonstration envelope	Below 9 200 ft	LIM-AFS-20-10
Automatic landing minimum weight, Automatic landing demonstration envelope	Above 256k lb	LIM-AFS-20-10
Automatic landing maximum weight	Below the maximum landing weight; up to 229 t (504 858 lb) at flight crew discretion	LIM-AFS-20-10
Terrain discontinuity evaluation distance for CAT I autoland	Within 300 m (974 ft) before the runway threshold	LIM-AFS-20-10
Automatic landing and rollout with outdated MAGVAR	Not permitted when the airport magnetic variation and the ADIRU magnetic variation differ by more than 3 °	AFM LIM-22-FGS
Minimum AP engagement height at takeoff	100 ft AGL and at least 5 s after liftoff	A330F LIM-AFS-10-10
Minimum AP use height in approach with F-G/S mode	200 ft AGL	A330F LIM-AFS-10-10
Minimum AP use height in approach with FINAL APP, V/S or FPA mode	250 ft AGL	A330F LIM-AFS-10-10
Minimum AP use height in circling approach	500 ft AGL	A330F LIM-AFS-10-10
Minimum AP use height, ILS approach with CAT1 on FMA, ILS approach when CAT1 is displayed on the FMA	160 ft AGL	A330F LIM-AFS-10-10
Minimum AP use height, GLS approach without AUTOLAND on FMA, GLS approach when AUTOLAND is not displayed on the FMA	160 ft AGL	A330F LIM-AFS-10-10
Minimum AP use height, LPV approach with SLS when CAT1 on FMA, LPV approach (with SLS) when CAT1 is displayed on the FMA	160 ft AGL	A330F LIM-AFS-10-10
Minimum AP use height, ILS approach with CAT2 or CAT3 on FMA	0 ft AGL if autoland	A330F LIM-AFS-10-10
Minimum AP engagement height after a manual go-around	100 ft AGL	A330F LIM-AFS-10-10
Minimum AP use height in all other phases	500 ft AGL	A330F LIM-AFS-10-10
AP/FD in OP DES or DES mode in approach, Approach using OP DES or DES mode	FCU selected altitude set to or above the higher of MDA/MDH or 500 ft AGL	A330F LIM-AFS-10-10
RNP AR achievable level	Could be below 0.3 NM, depending on the RNP AR application (refer to ACD)	A330F LIM-AFS-10-20
ILS CAT II minimum decision height	100 ft	A330F LIM-AFS-20
AP disconnect height for manual landing, ILS CAT II, ILS CAT II manual landing	No later than 80 ft AGL	A330F LIM-AFS-20
SA CAT II minimum decision height	100 ft	A330F LIM-AFS-20
AP disconnect height, SA CAT II automatic approach without autoland	No later than 80 ft AGL	A330F LIM-AFS-20
OTS CAT II minimum decision height	100 ft	A330F LIM-AFS-20
AP disconnect height, OTS CAT II automatic approach without autoland	No later than 80 ft AGL	A330F LIM-AFS-20
ILS CAT III fail passive (single) minimum decision height	50 ft	A330F LIM-AFS-20
ILS CAT III fail operational (dual) alert height	200 ft	A330F LIM-AFS-20
Minimum runway visual range, CAT III without DH	300 ft	A330F LIM-AFS-20
CAT II / CAT III autoland with engine out, Engine-out CAT II/CAT III autoland	Approved only in Configuration 3, engine-out procedures completed before 1 000 ft in approach	A330F LIM-AFS-20
Maximum headwind for ILS CAT II/CAT III and GLS CAT I	35 kt	A330F LIM-AFS-20
Maximum tailwind for ILS CAT II/CAT III and GLS CAT I	10 kt	A330F LIM-AFS-20

Limitation	Value	Ref
Maximum crosswind for ILS CAT II/CAT III and GLS CAT I	15 kt	A330F LIM-AFS-20
Automatic landing approved configurations, ILS CAT II/CAT III autoland and GLS CAT I autoland	CONF 3 and CONF FULL	A330F LIM-AFS-20-10
Automatic landing demonstrated glide slope angle, Automatic landing demonstration envelope	Between -2.5 ° and -3.25 °	A330F LIM-AFS-20-10
Automatic landing maximum airfield elevation, Automatic landing demonstration envelope	Below 9 200 ft	A330F LIM-AFS-20-10
Automatic landing minimum weight, Automatic landing demonstration envelope	Above 256k lb	A330F LIM-AFS-20-10
Automatic landing maximum weight	Below the maximum landing weight; up to 229 t (504 858 lb) at flight crew discretion	A330F LIM-AFS-20-10
Terrain discontinuity evaluation distance for CAT I autoland	Within 300 m (974 ft) before the runway threshold	A330F LIM-AFS-20-10
Engines		
EGT limit, takeoff and go-around	920 °C for 20 s / 900 °C for 5 min all engines operative, 10 min one engine inoperative	LIM-ENG
EGT limit, Maximum Continuous Thrust (MCT)	850 °C, not limited in time	LIM-ENG
EGT limit, starting on ground, Engine start on ground	700 °C	LIM-ENG
EGT limit, starting in flight, Engine start in flight	850 °C	LIM-ENG
Maximum N1	99 %	LIM-ENG
Maximum N2	103.3 %	LIM-ENG
Maximum N3	100 %	LIM-ENG
Maximum continuous oil temperature	190 °C	LIM-ENG
Minimum oil temperature before takeoff	50 °C	LIM-ENG
Minimum oil quantity, Dispatch / preflight	Highest of 15 qt, or 6 qt + estimated consumption (average estimated consumption 0.7 qt/h)	LIM-ENG
Minimum oil pressure	25 PSI	LIM-ENG
Starter maximum continuous operation	5 min	LIM-ENG
Starter duty cycles, Engine start	Two 3 min cycles and a consecutive 1 min cycle, with run down to zero N3 between each cycle	LIM-ENG
Starter cooling period, After one continuous operation or the three cycles	30 min	LIM-ENG
Maximum N3 for running engagement of the starter	10 % on ground, 30 % in flight	LIM-ENG
Minimum speed for maximum reverse thrust	70 kt (idle reverse permitted down to aircraft stop)	LIM-ENG
Maximum flexible temperature (TMAXFLEX)	ISA + 60 °C	LIM-ENG-10
Engine crosswind limit at takeoff, Takeoff, ground operation	32 kt (gust included)	LIM-ENG
EGT limit, takeoff and go-around	920 °C for 20 s / 900 °C for 5 min all engines operative, 10 min one engine inoperative	A330F LIM-ENG
EGT limit, Maximum Continuous Thrust (MCT)	850 °C, not limited in time	A330F LIM-ENG
EGT limit, starting on ground, Engine start on ground	700 °C	A330F LIM-ENG
EGT limit, starting in flight, Engine start in flight	850 °C	A330F LIM-ENG
Maximum N1	99 %	A330F LIM-ENG
Maximum N2	103.3 %	A330F LIM-ENG
Maximum N3	100 %	A330F LIM-ENG
Maximum continuous oil temperature	190 °C	A330F LIM-ENG
Minimum oil temperature before takeoff	50 °C	A330F LIM-ENG
Minimum oil quantity, Dispatch / preflight	Highest of 15 qt, or 6 qt + estimated consumption (average estimated consumption 0.7 qt/h)	A330F LIM-ENG
Minimum oil pressure	25 PSI	A330F LIM-ENG
Starter maximum continuous operation	5 min	A330F LIM-ENG
Starter duty cycles, Engine start	Two 3 min cycles and a consecutive 1 min cycle, with run down to zero N3 between each cycle	A330F LIM-ENG
Starter cooling period, After one continuous operation or the three cycles	30 min	A330F LIM-ENG
Maximum N3 for running engagement of the starter	10 % on ground, 30 % in flight	A330F LIM-ENG
Minimum speed for maximum reverse thrust	70 kt (idle reverse permitted down to aircraft stop)	A330F LIM-ENG
Maximum flexible temperature (TMAXFLEX)	ISA + 60 °C	A330F LIM-ENG-10

Limitation	Value	Ref
Engine crosswind limit at takeoff, Takeoff, ground operation	32 kt (gust included)	A330F LIM-ENG
Fire Protection / ETOPS		
Cargo fire suppression system time capability, ETOPS significant-system time capability (planning)	260 min	AFM APP-ETOPS-0000553 9.0002001 (ETOPS Supplement - LIMITATIONS)
Flight Controls		
Flight maneuvering load acceleration limit, Clean configuration	-1 g to +2.5 g	LIM-AG-F_CTL
Flight maneuvering load acceleration limit, Other configurations (slats and/or flaps extended)	0 g to +2 g	LIM-AG-F_CTL
Maximum operating altitude with slats and/or flaps extended	20 000 ft	LIM-F_CTL
Flight maneuvering load acceleration limit, Clean configuration	-1 g to +2.5 g	A330F LIM-AG-F_CTL
Flight maneuvering load acceleration limit, Other configurations (slats and/or flaps extended)	0 g to +2 g	A330F LIM-AG-F_CTL
Maximum operating altitude with slats and/or flaps extended	20 000 ft	A330F LIM-F_CTL
Fuel		
Certified fuels	JET A, JET A1, JET B, JP4, JP5, JP8, N°3 JET, RT and TS-1	LIM-FUEL
Maximum fuel temperature, JET A, JET A1, JP5, JP8, N°3 JET, RT and TS-1	+55 °C	LIM-FUEL
Maximum fuel temperature, JP4, JET B	+49 °C	LIM-FUEL
Minimum fuel temperature, inner tank, Inner tank; in outer or trim tank the limit is the freezing point	Highest of freezing point or -44 °C below 30 000 ft, or -54 °C above 30 000 ft	LIM-FUEL
Maximum allowed wing fuel imbalance, inner tanks (outer balanced)	6k lb at full; 11k lb at 37k lb; 17k lb at 17k lb	LIM-FUEL
Maximum allowed wing fuel imbalance, outer tanks (inner balanced)	3k lb at full; 5k lb / 3k lb; 4k lb at 4k lb	LIM-FUEL
Fuel imbalance below which no limitation applies, Variation is linear between the tabulated values	17k lb inner tanks, 4k lb outer tanks	LIM-FUEL
Minimum fuel quantity for takeoff, Takeoff; fuel wing tank low level ECAM alerts must not be present	11k lb	LIM-FUEL
Certified fuels	JET A, JET A1, JET B, JP4, JP5, JP8, N°3 JET, RT and TS-1	A330F LIM-FUEL
Maximum fuel temperature, JET A, JET A1, JP5, JP8, N°3 JET, RT and TS-1	+55 °C	A330F LIM-FUEL
Maximum fuel temperature, JP4, JET B	+49 °C	A330F LIM-FUEL
Minimum fuel temperature, inner tank, Inner tank; in outer or trim tank the limit is the freezing point	Highest of freezing point or -44 °C below 30 000 ft, or -54 °C above 30 000 ft	A330F LIM-FUEL
Maximum allowed wing fuel imbalance, inner tanks (outer balanced)	6k lb at full; 11k lb at 37k lb; 17k lb at 17k lb	A330F LIM-FUEL
Maximum allowed wing fuel imbalance, outer tanks (inner balanced)	3k lb at full; 5k lb / 3k lb; 4k lb at 4k lb	A330F LIM-FUEL
Fuel imbalance below which no limitation applies, Variation is linear between the tabulated values	17k lb inner tanks, 4k lb outer tanks	A330F LIM-FUEL
Minimum fuel quantity for takeoff, Takeoff; fuel wing tank low level ECAM alerts must not be present	11k lb	A330F LIM-FUEL
Ice and Rain		
Icing conditions - temperature threshold	OAT (on ground or after takeoff) or TAT (in flight) at or below 10 °C with visible moisture present	LIM-ICE_RAIN
Icing conditions - ground contamination	OAT on the ground and for takeoff at or below 10 °C when operating on surfaces with snow, standing water or slush	LIM-ICE_RAIN
Severe ice accretion threshold, Ice accumulation on the airframe	Approximately 5 mm (0.2 in) thick or more	LIM-ICE_RAIN
Engine anti-ice, operations in icing conditions	ON for all ground and flight operations when icing conditions exist or are anticipated, except during climb and cruise below -40 °C SAT; ON prior to and during descent in icing conditions, including below -40 °C SAT	AFM LIM-70
Icing conditions - temperature threshold	OAT (on ground or after takeoff) or TAT (in flight) at or below 10 °C with visible moisture present	A330F LIM-ICE_RAIN
Icing conditions - ground contamination	OAT on the ground and for takeoff at or below 10 °C when operating on surfaces with snow, standing water or slush	A330F LIM-ICE_RAIN

Limitation	Value	Ref
Severe ice accretion threshold, Ice accumulation on the airframe	Approximately 5 mm (0.2 in) thick or more	A330F LIM-ICE_RAIN
Landing Gear		
Maximum brake temperature for takeoff, Brake fans off	300 °C	LIM-LG
Speed during a turn using asymmetric thrust, Ground turn with asymmetric thrust	Between 5 and 10 kt	LIM-LG-10
Towbarless operation on the nose landing gear, Towing and pushback	Approved with accepted towbarless towing vehicles listed in Airbus WISE ISI 09.11.00001	LIM-LG-10
Maximum taxi speed during turn with deflated tire(s)	7 kt	LIM-LG
Maximum taxi speed with two tires deflated on the same main gear, Two tires deflated on the same main gear (maximum one main gear)	3 kt	LIM-LG
Maximum NWS angle with two tires deflated on the same main gear, Two tires deflated on the same main gear (maximum one main gear)	30 °	LIM-LG
Maximum brake temperature for takeoff, Brake fans off	300 °C	A330F LIM-LG
Speed during a turn using asymmetric thrust, Ground turn with asymmetric thrust	Between 5 and 10 kt	A330F LIM-LG-10
Towbarless operation on the nose landing gear, Towing and pushback	Approved with accepted towbarless towing vehicles listed in Airbus WISE ISI 09.11.00001	A330F LIM-LG-10
Maximum taxi speed during turn with deflated tire(s)	7 kt	A330F LIM-LG
Maximum taxi speed with two tires deflated on the same main gear, Two tires deflated on the same main gear (maximum one main gear)	3 kt	A330F LIM-LG
Maximum NWS angle with two tires deflated on the same main gear, Two tires deflated on the same main gear (maximum one main gear)	30 °	A330F LIM-LG
Navigation		
ITP maximum altitude change, In-Trail Procedure climb or descent	4 000 ft	PRO-SPO-90-10
ITP minimum rate of climb	300 ft/min	PRO-SPO-90-10
ITP maximum reference aircraft	2 reference aircraft	PRO-SPO-90-10
IRS ground alignment latitude limits, IR alignment in NAV mode, on ground only	Between 82 ° North and 82 ° South	LIM-NAV
Latitudes where MAG heading/track is invalid (all ADIRUs same magvar table), NAV mode; TRUE reference must be selected beyond these limits	North of 73 ° North between 90 ° West and 120 ° West; north of 82 ° North; south of 60 ° South	LIM-NAV
Latitudes where MAG heading/track is invalid (one ADIRU with a different magvar table), NAV mode; flying beyond these limits is prohibited	North of 60 ° North between 30 ° West and 160 ° West; north of 75 ° North; south of 55 ° South	LIM-NAV
ITP maximum altitude change, In-Trail Procedure climb or descent	4 000 ft	A330F R10 (not located)
ITP minimum rate of climb	300 ft/min	A330F R10 (not located)
ITP maximum reference aircraft	2 reference aircraft	A330F R10 (not located)
IRS ground alignment latitude limits, IR alignment in NAV mode, on ground only	Between 82 ° North and 82 ° South	A330F LIM-NAV
Latitudes where MAG heading/track is invalid (all ADIRUs same magvar table), NAV mode; TRUE reference must be selected beyond these limits	North of 73 ° North between 90 ° West and 120 ° West; north of 82 ° North; south of 60 ° South	A330F LIM-NAV
Latitudes where MAG heading/track is invalid (one ADIRU with a different magvar table), NAV mode; flying beyond these limits is prohibited	North of 60 ° North between 30 ° West and 160 ° West; north of 75 ° North; south of 55 ° South	A330F LIM-NAV
Navigation / PBN (RNP 4 & RNP 10 oceanic)		
IRS-only navigation time limit (one GPS required beyond)	6.2 hr from IRS ground alignment / 5.7 hr from last FM position update	PRC p.39 'Oceanic References, Required RNP 10 Equipment' (card cites PRO-SPO Performance Based Navigation (PBN))
Operating Limitations / Environmental Envelope		
Maximum runway (airport) altitude, Takeoff and landing (environmental envelope)	12,500 ft	LIM-AG-OPS-ENV-000216 54.0013001 / AFM LIM-OP S-00005456.0003001
Oxygen		
Minimum flight crew oxygen bottle pressure	Tabulated by REF temperature -10 °C to 50 °C: 2 crew 520 to 640 PSI; 2 crew + 1 OBS 660 to 810 PSI; 2 crew + 2 OBS 810 to 1 000 PSI	LIM-OXY

Limitation	Value	Ref
Minimum bottle pressure design case (not a crew limitation): protection during emergency descent	15 min for all flight crewmembers and observers	LIM-OXY
Minimum bottle pressure design case (not a crew limitation): protection during cruise at FL 100	105 min for two crewmembers	LIM-OXY
Minimum bottle pressure design case (not a crew limitation): protection against smoke, 100 % oxygen; basis for the minimum bottle pressure	15 min at 8 000 ft cabin altitude for all flight crewmembers and observers	LIM-OXY
Minimum flight crew oxygen bottle pressure	Tabulated by REF temperature -10 °C to 50 °C: 2 crew 520 to 640 PSI; 2 crew + 1 OBS 660 to 810 PSI; 2 crew + 2 OBS 810 to 1 000 PSI	A330F TDU
Minimum bottle pressure design case (not a crew limitation): protection during emergency descent	15 min for all flight crewmembers and observers	A330F LIM-OXY-90000021.9000120
Minimum bottle pressure design case (not a crew limitation): protection during cruise at FL 100	105 min for two crewmembers	A330F R10 (not located)
Minimum bottle pressure design case (not a crew limitation): protection against smoke, 100 % oxygen; basis for the minimum bottle pressure	15 min at 8 000 ft cabin altitude for all flight crewmembers and observers	A330F TDU
Reference (AFM procedures)		
Minimum RAT speed, RAT deployed (EMER ELEC CONFIG; also G+B SYS LO PR procedure)	140 kt	AFM EMER-24-00005218.0006001 (ELEC - EMER CONFIG); also AFM EMER-29-00005726.0006001 (HYD - G+B SYS LO PR): 'Minimum RAT speed: 140 kt.'
APU battery start in flight (APU TR not available), APU start in flight with APU TR not available	Below 25 000 ft	AFM NORM-49-00005815.0002001 (Auxiliary Power Unit - Starting In Flight)
Speeds		
Maximum operating speed (VMO)	330 kt IAS	AFM LIM-SPD
Maximum operating Mach (MMO)	M 0.86	AFM LIM-SPD
Cockpit window open maximum speed	230 kt	LIM-AG-SPD
Maximum flaps/slats speed - CONF 1 (slats 1), Flaps lever 1, holding; unit 'kt' corroborated by the AFM VFE table	240 kt	LIM-AG-SPD
Maximum flaps/slats speed - CONF 1+F, Flaps lever 1, takeoff; unit 'kt' corroborated by the AFM VFE table	215 kt	LIM-AG-SPD
Maximum flaps/slats speed - CONF 2	196 kt	LIM-AG-SPD
Maximum flaps/slats speed - CONF 3	186 kt	LIM-AG-SPD
Maximum flaps/slats speed - CONF FULL	180 kt	LIM-AG-SPD
Maximum speed with landing gear extended (VLE)	250 kt/0.55	LIM-AG-SPD
Maximum speed with landing gear operating (VLO), Landing gear extension and retraction	250 kt/0.55	LIM-AG-SPD
Maximum speed for landing gear gravity extension (VLE/VLO)	200 kt	LIM-AG-SPD
Maximum tire speed (ground speed)	204 kt	LIM-AG-SPD
Minimum control speed for landing (VMCL)	118 kt (IAS)	LIM-AG-SPD
Wipers maximum operating speed, Wipers operating	230 kt	LIM-AG-SPD
Cockpit window open maximum speed	230 kt	A330F DSC-22_10-50-30
Maximum flaps/slats speed - CONF 1 (slats 1), Flaps lever 1, holding; unit 'kt' corroborated by the AFM VFE table	240 kt	A330F LIM-AG-SPD
Maximum flaps/slats speed - CONF 1+F, Flaps lever 1, takeoff; unit 'kt' corroborated by the AFM VFE table	215 kt	A330F LIM-AG-SPD
Maximum flaps/slats speed - CONF 2	196 kt	A330F LIM-AG-SPD
Maximum flaps/slats speed - CONF 3	186 kt	A330F LIM-AG-SPD
Maximum flaps/slats speed - CONF FULL	180 kt	A330F LIM-AG-SPD
Maximum speed with landing gear extended (VLE)	250 kt/0.55	A330F LIM-AG-SPD
Maximum speed with landing gear operating (VLO), Landing gear extension and retraction	250 kt/0.55	A330F LIM-AG-SPD
Maximum speed for landing gear gravity extension (VLE/VLO)	200 kt	A330F LIM-AG-SPD
Maximum tire speed (ground speed)	204 kt	A330F LIM-AG-SPD

Limitation	Value	Ref
Minimum control speed for landing (VMCL)	126 kt	A330F LIM-AG-SPD
Wipers maximum operating speed, Wipers operating	230 kt	A330F LIM-AG-SPD
Surveillance		
Distance for inhibiting predictive GPWS (TERR pb OFF)	Less than 15 NM from the airfield	LIM-SURV
Distance for inhibiting predictive GPWS (TERR pb OFF)	Less than 15 NM from the airfield	A330F LIM-SURV
Weights		
Maximum taxi weight, A330 PAX fleet	527k lb	LIM-AG-WGHT
Maximum takeoff weight (brake release), A330 PAX fleet	525k lb	LIM-AG-WGHT
Maximum landing weight, A330 PAX fleet	401k lb	LIM-AG-WGHT
Maximum zero fuel weight	from 370k lb to 375k lb	LIM-AG-WGHT
Minimum weight, A330 PAX fleet	256k lb	LIM-AG-WGHT
Maximum taxi weight, A330 freighter fleet	516k lb	A330F LIM-AG-WGHT
Maximum takeoff weight (brake release), A330 freighter fleet	514k lb	A330F LIM-AG-WGHT
Maximum landing weight, A330 freighter fleet	412k lb	A330F LIM-AG-WGHT
Maximum zero fuel weight	386k lb	A330F LIM-AG-WGHT
Minimum weight, A330 freighter fleet	267k lb	A330F LIM-AG-WGHT